

A numerical investigation of fan wall cooling system for modular air-cooled data center

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ABSTRACT

Airflow management in the data center is one of the most researched topics to achieve uniform airflow for effective thermal management. Recently, the Fan Wall Cooling (FWC) system is popular in air-cooled data centers due to the advantage of no raised floor plenum. In this research, a validated CFD model is used to study the FWC performance with different configurations to gain insights for the optimum design. Five new configurations are proposed to optimize the FWC performance along with ordinary FWC. Each of them has a different rack-tilted and row-tilted angle. The results show that tilting the rack by 15° towards the supply air direction can increase the rack air flowrate by 2.5% (800 CFM), and reduce the maximum normalized server temperature ($T_{ser,max}^*$) by 7% from 1.30 to 1.21. The CFD simulations also reveal that all new configurations can effectively mitigate the airflow maldistribution. Lastly, the FWC exhibits superior thermal performance compared to the conventional Underfloor Air Delivery (UFAD) and the Overhead Air Supply (OHAS) methods.

1. Introduction

The virus pandemic in 2020 and emerging 5G telecommunication technology dramatically boost the demand for high-quality data transmission and processing services such as web conferences, virtual workshops, remote working etc. Data center (DC), the infrastructure accommodating thousands of information technology equipment (ITE), has become more critical than ever before. Koot et al. [1] forecasted the global DC electricity needs would increase from 286 TWh in 2016 to 321 TWh in 2030 if all known growth factors remain the same. The report from Uptime Institute [2] shows that the average Power Usage Effectiveness (PUE) [3] of 300 data centers worldwide struggling to reduce in the past five years from 2015 to 2020, suggesting the lack of the new energy saving technology to support the industry. To ensure the reliable operations and longevity of servers and IT equipment that are housed in DC, its thermal environment has to be maintained within specific recommended temperature and humidity ranges. As most DCs are air-cooled and the fact that air is a poor heat transfer medium, this has led to 30 to 40% of its total energy usage going towards cooling [4]. Advanced cooling solutions such as liquid-cooling [5], immersion cooling, etc. [6,7] provide opportunities to cut the cooling electrical power dramatically. However, the concerns on the leakage [8], recent

advances in air-cooled heat sink [9], the significant energy saving obtained from reduced air flowrate due to implementation of the thermosyphon heat sinks [10] and smart fan controls [11] make the air cooling method still the primary choice of DC's cooling system. Therefore, it is necessary to improve the airflow distribution of the data centers.

Managing the airflow in the data center provides opportunities to save fan power and elevate the cooling setpoint. An ideal airflow loop in the data center should be a single circuit with all the produced cold air entering the servers and exhausted hot air traveling back to the heat exchanger. However, the presence of the cold air bypass [12] and hot air recirculation [13] degrade the cooling efficiency. Many studies have been conducted to improve the air loop in the data center. Phan et al. [14] analyzed the perforated tile design in the underfloor air delivery (UFAD) to reduce the cold air bypass and the influence of turbulence on the modeling accuracy. Experiments conducted by Khalili et al. [15] revealed the excellent flow strengthening effect from the low perforation tile. However, the measurements conducted by Arghode et al. [16] showed excessive bypass air attributing to the high-velocity jet. Nada et al. [17] found a similar issue with the inappropriate supply air direction and emphasize allocating the CRAC unit perpendicular to the cold aisle. Back in 2003, Karki et al. [18] proposed a list of techniques to control the tile airflow. VanGilder et al. [19] suggested using the low

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Nomenclature

AFC	Active Fan Curve
CFD	Computational Fluid Dynamic
CRAC	Computer Room Air Conditioner
dp	Pressure difference
FWC	Fan Wall Cooling
UFAD	Underfloor Air Delivery
TDP	Thermal Design Power
T	temperature ($^{\circ}\text{C}$)
P	pressure (Pa)
V	volumetric flowrate (m^3/s)
m	mass flow rate (kg/s)
u	velocity (m/s)
κ	thermal conductivity ($\text{W}/\text{m K}$)
k	turbulent kinetic energy
ϵ	turbulent dissipation
E	power input (W)
TC	thermocouple
S	momentum source term
BF	body-force model
FD	free delivery

TC	thermocouple
rap	return air plenum

Greek symbols

ρ	density (kg/m^3)
μ	dynamic viscosity (kg/ms)
Φ	maldistribution factor (%)
$\forall$	volume (m^3)
α	rack-tilted angle ($^{\circ}$)
β	row-tilted angle ($^{\circ}$)

Subscripts

0	Free Delivery Point
ser	server
a	air
bcf	backflow
in	inlet
out	outlet
high	high server load
LB	Load Bank
BF	Body-Force

perforation tile for a better airflow uniformity. Khalili et al. [20] then discovered that the bottom servers (measured from 0 to 8U starting from the bottom of the rack) generally have 3 $^{\circ}\text{C}$ –4.5 $^{\circ}\text{C}$ higher inlet air temperature due to hot air recirculation induced by tile air jet from under-rack leakage gap. Tawackolian et al. [21] showed the high-velocity air jet created by perforated tile induced excessive hot air recirculation to servers located at the bottom of the rack, and increased their power consumption by 6%. The low perforation tile has the inherent massive pressure loss and low airflow rate. Joshi et al. [22] highlighted the pressure loss due to the raised floor plenum could be 150 Pa. Silva-Llanca et al. [23] showed 34% of the total exergy loss (exergy stands for the availability of air to do positive work) at air the delivery path to the raised floor plenum, suggesting the chilled water pipe, electrical cable and fire fighting device etc. contribute significant inefficiency to the cooling system. All these drawbacks increase the CRAC fan size and power consumption. Yuan et al. [24] introduced an in-rack cold aisle containment to eliminate the cold air bypass and achieved 98 kWh energy saving per day. Their subsequent study [25] utilizing a tilted server placement in the rack reduced the hot spot by 3.3 $^{\circ}\text{C}$. Lim et al. [13] suggested a similar server placement angle by adjusting the server exhaust angle to eliminate the hot spot. Song et al. [26] used a convex deflector in the cold aisle to reduce the cold air bypass and greatly improved the airflow uniformity. Ahmadi et al. [27] proposed to purposely bypass a stream of the return air from the CRAC

to reduce the fan power. This is true due to the high pressure drop from the heat exchanger and filter in the CRAC. Lu et al. [28] proposed a raised floor plenum with a gradient cross-section and reduced the hot spot by 2.9 $^{\circ}\text{C}$. Similarly, Zhang et al. [29] proposed the T-shape underfloor air duct to even the pressure distribution in the plenum. Fulpagare et al. [30] proposed a lightweight forecast model to predict the thermal behaviour of raised floor plenum data center, the experimental results revealed it was capable of providing 10 min lead time to prepare the cooling resource for hot spot mitigation.

Due to excessive thermal inefficiencies in traditional UFAD cooling, some alternative solutions (in-row cooler [31], overhead cooler [32], evaporative cooler [33], etc.) were proposed to get rid of the raised floor plenum and eliminate the possible high-velocity air jet. Cho et al. [34] showed experimentally that the in-row cooler was 37.1% better in Supply Heat Index (SHI) than UFAD cooling. Gupta et al. [32] compared four different cooling architectures. The proposed rack-mounted cooler reduced the exergy loss by 15%. Hosein et al. [35] found that the in-row cooler can save 29% more energy than the traditional UFAD. Nada et al. [36] investigated three possible containments for rack modules equipped with in-row coolers. They managed to reduce the hot spot by 5 $^{\circ}\text{C}$. Huang et al. [37] provide a comprehensive review, summarizing the pros and cons of various rack-level cooling solutions. Jin et al. [38] found that the retrofitted in-row cooling layout from the traditional UFAD reduced the SHI from 0.79 to 0.31 and could possible to save 110,

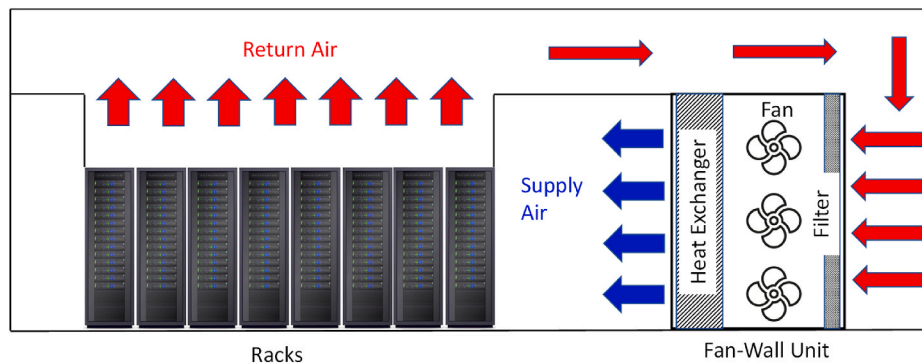


Fig. 1. Fan-Wall Cooling (FWC) concept.

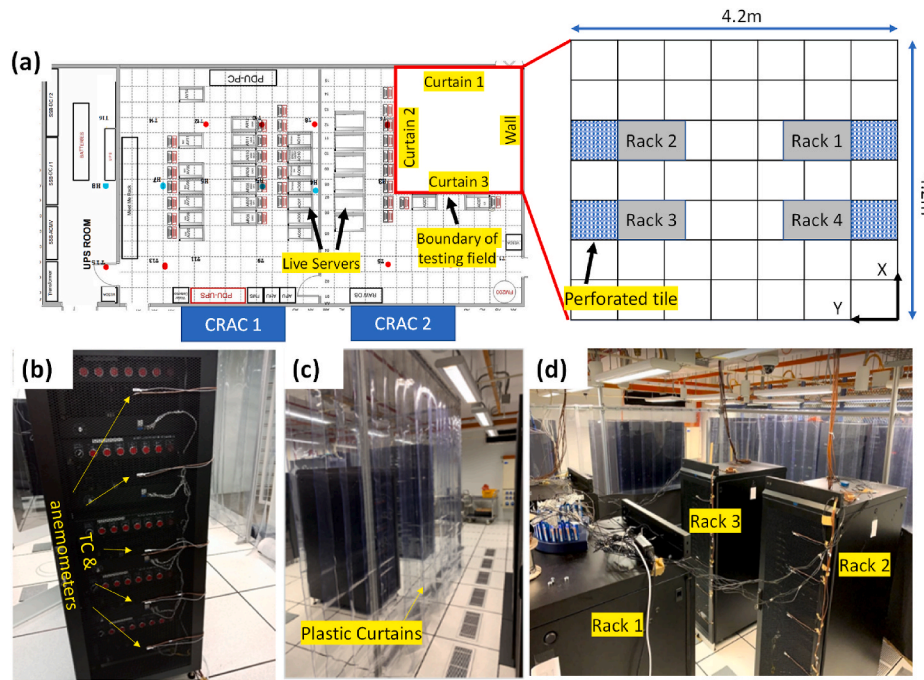


Fig. 2. (a) Location of the testing field in the existing server room; (b) Front view of 32U rack; (c) External view of the test field with curtain insight; (d) Internal view of the test zone.

829 kWh per year. The inherent shortened air path for either the in-row or rack-mounted cooler improves the cooling efficiency. However, Nemati et al. [39] found that the flow maldistribution in an in-row cooler system could be more severe than UFAD and created hot spot in the cold aisle. Alisa et al. [40] highlighted that the in-row cooler's low thermal mass would significantly reduce the lead time when a cooling failure occurs, which gives less time to initiate the back-up system. Thus, a cooling system providing a better air distribution, less pressure loss and a larger thermal mass is highly desired. Chu et al. [41] studied the Overhead Air Supply (OHAS) in a container data center. Their study showed that the OHAS is more spatially efficient than UFAD, and it can provide a good thermal environment with proper design on ceiling grill size and deflector.

Recently the fan-wall cooling (FWC) system was proposed [42]. It is an array of the fan mounted on the data center's perimeter wall that supplies the air from the side, as illustrated in Fig. 1. It is commonly used

in the data center with the water-side economizer [43]. Since it eliminates the need for the raised floor plenum, it is cheaper than the UFAD and has a lower room height requirement. Moreover, the FWC is highly modularized, offers excellent flexibility for expansion or downsizing. Currently, there are limited literature on the FWC system and none of them provided a guideline for data center designers. In this study, a validated CFD is adopted to characterize the FWC performance. The influence of FWC cold aisle and hot aisle widths on the flow maldistribution and rack inlet air temperature is investigated. The data center designers could refer to these findings to improve the FWC layout designs.

This study aims to: (1) compares the flow distribution, server temperature performance among the standard UFAD, FWC and overhead air supply (OHAS); (2) to investigate the influence of the cold and hot aisle size on the FWC performance, so that to evaluate the possibility to shrink the DC size; (3) to optimize the FWC layout by introducing rack-tilted and row-tilted configurations to mitigate the flow maldistribution, reduce hot air recirculation, and improve the overall performance of the FWC thermal environment.

This paper is organized as follows: Section 2 discusses the method to conduct the validation and numerical scheme. Section 3.1 first establishes a validated CFD model by field measurements conducted in an existing server room. The validated model is then modified and transferred to Section 3.2 to compare the FWC performance with the conventional UFAD and overhead air supply (OHAS) cooling method. Section 3.3 discusses the characteristics of the FWC by looking at the influence of the cold aisle width (S) and hot aisle width (H). Section 3.4 discusses the optimization of the FWC with rack-tilted and row-tilted configurations.

2. Model and method

2.1. Model validation

As it is too costly to set up a dedicated experimental facility to test FWC, a numerical model is used to study its physics. Field measurements were conducted to validate the model to ensure its accuracy. Since the

Table 1
Summary of the testing condition, equipment and sensors.

Items	Dimension	Remark
Test Zone	4.2 m (L) x 4.2 m (W)	Enveloped by 2.5 m high plastic curtains
Rack	1.0 m (L) x 0.6 m (W) x 1.53 m (H)	32U rack, equipped with 5 load banks. 2.5 kW/rack for measurement
Load Bank	0.6 m (L) x 0.45 m (W) x 0.264 m (H)	6U, 0.5 kW/load bank, airflow rate is set to 94 CFM to produce 9.8 °C delta T
Perforated Tiles	0.6 m (L) x 0.6 m (W)	Raised floor plenum is at 15 Pa static, 17 °C temperature. Tile airflow rate = 507 CFM
Miscellaneous	–	6 LED lighting, each 10 W. Two cable trays assumed with no heat generation.
Temperature Sensor	–	T-Type thermocouple, ± 0.5 °C
Anemometer	–	Hot-wire anemometer DegreeC (UAS1000), ± 0.025 m/s
Differential Pressure Transducer	–	OMEGA PX653-0.5D5V with 0.5 "WC range

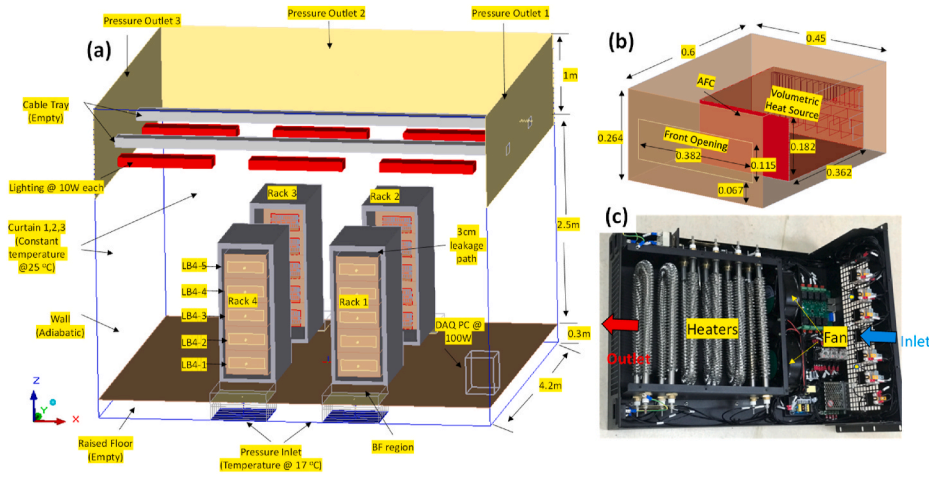


Fig. 3. (a) CFD model for 4-rack field validation; (b) CFD model for load bank; (c) Load bank's interior.

FWC eliminates the raised floor plenum and perforated tile, the boundary conditions are more straightforward and can be directly transferred from the validated model. The field measurements were conducted in a live server room on the National University of Singapore campus, as shown in Fig. 2. Four 32U racks, each installed with five 6U load banks, were deployed. Load banks are usually used in unoccupied DC to simulate the server heat load to test the facility. Plastic curtains of 2.5 m height were installed to isolate the testing field to minimize the infiltration of hot air from the other operating servers into the test area. A top clearance of 1 m has to be maintained to comply with the fire safety requirement. The server room has two CRACs supplying 17 °C cold air through the 600 mm deep raised floor plenum. The 6U load bank and perforated tile were separately characterized in a flow bench wind tunnel. The rack's heating power was set to 2.5 kW/rack due to the limited cooling capacity of the CRAC in the server room. The airflow rate was set to produce 9.8 °C temperature difference before and after the load bank, referred from a 1U server [44,45]. The actual server temperature difference varies with the server utilization level. The 9.8 °C is the median point of the reference server in-between the idle and full load status. Therefore, it is selected for the experiment, the corresponding reference server air flowrate at 9.8 °C temperature difference is 94 CFM. Table 1 provides a summary of the testing conditions, equipment specifications, and sensor information.

Ansys Icepak 19.0 was used to create the numerical model. The load bank was constructed with a compact modeling approach which consists of a front opening, an internal fan and a volumetric heat source. The interior details (PCB boards, fin on the heater, heater rods) are ignored to reduce the mesh elements. The active fan curve (AFC) is measured in the flow bench and assigned to the internal fan. The rack was modeled as the empty object and assumed to be adiabatic. A 3 cm wide leakage path is created to allow the by-pass or recirculation to occur, as shown in Fig. 3. The test zone was modeled as a 4.2 m (L) x 4.2 m (W) x 3.5 m (H) volume. Curtains 1 to 3 were modeled as no-slip wall with 25 °C constant wall temperature determined by measuring the ambient temperature outside the test zone. The 1 m opening above the curtain was modeled as the pressure outlet with 25 °C backflow temperature. A negative pressure (−1 Pa static) was assigned to pressure outlets 2 and 3 to simulate the return air towards CRAC. While pressure outlet 1 was assigned with 0 Pa static. The actual measurement revealed the difficulties of quantifying the outlet pressure value due to the large opening size and the disruptions from live servers outside. Four pressure inlets with 15 Pa and temperature 17 °C were located 0.3 m underneath the perforated tile. The inlet pressure value was obtained from the pressure transducer, which measured the pressure difference between the raised floor plenum and the room. The perforated tile was modeled as a porous jump surface with a pressure drop coefficient. The pressure drop of the

Table 2
Boundary Conditions for model validation.

Boundary	Inputs
Pressure Inlet	$P_{in} = 15 \text{ Pa}$ $T_{a,s} = 17 \text{ °C}$
Pressure Outlet 1	$P_{out} = 0 \text{ Pa}$ $T_{a,bcf} = 25 \text{ °C}$
Pressure Outlets 2, 3	$P_{out} = -1 \text{ Pa}$ $T_{a,bcf} = 25 \text{ °C}$
Perforated Tile (Porous Jump Surface)	$C_{tile} = 13.57$
Body-Force Region	$\forall = 0.6 \text{ m} \times 0.6 \text{ m} \times 0.1 \text{ m}$ $S_{BF,z} = \frac{C_{BF} V_{app}}{\forall}, C_{BF} = 22$
Load Bank Internal Fan	$V_{LB,low,0} = 94 \text{ CFM}$ Active Fan Curve*: $\Delta P_{LB,low} = -0.177 V_{LB,low} + 16.642$
Heater	Volumetric Heat Source $E_{LB} = 500 \text{ W}$ (10% of 500 W is assigned to LB chassis surface)
Curtain 1 - 3	Constant wall temperature at 25 °C
Wall, Cable Tray	Adiabatic
Lighting, DAQ PC	Constant heat flux

*Active Fan Curve is a combined impedance and fan curve. More detail can be referred in the study [47].

tile was calculated using equation (1). A body-force (BF) region with a momentum source was imposed above the perforated tile to simulate the air jet effect. The BF model to simulate the perforated tile jet can be referred from Ref. [46]. The summary of the boundary conditions for the validation model is listed in Table 2.

$$\Delta P_{tile} = C_{tile} \frac{1}{2} \rho \bar{u}_{tile}^2 \quad (1)$$

where P_{in} and P_{out} are the inlet and outlet pressure in Pa, $T_{a,s}$ and $T_{a,bcf}$ are the supply and backflow air temperature in °C. C_{tile} and C_{BF} are the perforated tile pressure drop and Body-Force (BF) coefficient (dimensionless). $S_{BF,z}$ is the momentum source term defined in BF region to simulate air jet from perforated tile. $V_{LB,low,0}$ is the free-delivery load bank air flowrate in CFM at low-speed setting. E_{LB} is the LB input power in W.

2.2. Design of the fan-wall cooling

The calibrated CFD model is used to simulate the FWC system with the following modifications: (1) The rack's height increases to 42U (2 m) with 7 load banks per rack since 42U rack is the most commonly used in the data center; b) the load banks (LB) heat load and fan speed adjusted

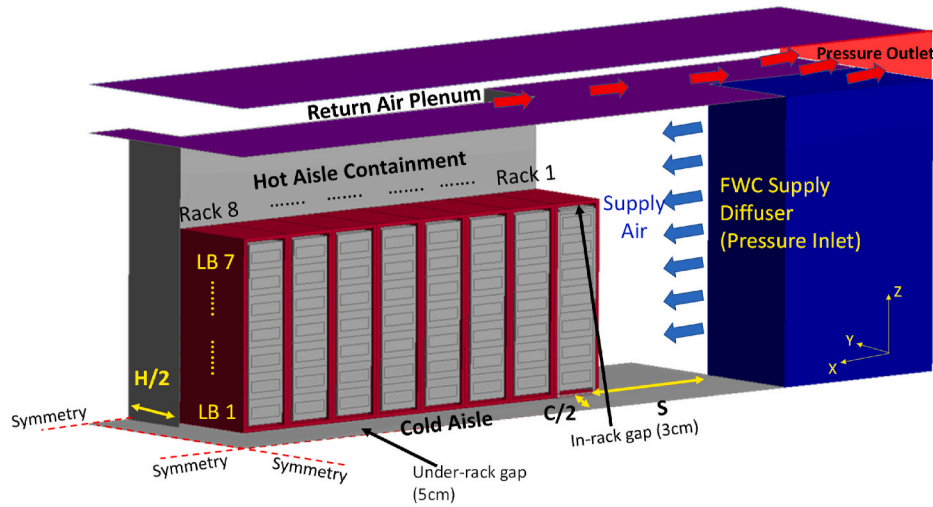


Fig. 4. FWC numerical model (half domain).

Table 3
Boundary conditions for FWC.

Boundary	Inputs
Pressure Inlet (FWC Supply Diffuser)	$P_{in} = 70 \text{ Pa}$ $T_{a.s} = 25^\circ \text{C}$ Dimension: $1.2 \text{ m} \times 3 \text{ m}$
Pressure Outlet	$P_{out} = -60 \text{ Pa}$ Dimension: $2.4 \text{ m} \times 0.5 \text{ m}$
Load Bank Internal Fan	High Server Load: $V_{LB,high,0} = 261 \text{ CFM}$ Active Fan Curve: $\Delta P_{LB,high} = -0.4293V_{LB,high} + 112.05$
Load Bank Heater (1 rack consists of 7 load banks)	High Server Load: $E_{LB,high} = 1400 \text{ W}$ (10% of 1400 W is assigned to LB chassis surface)
Return Air Plenum Pressure Drop Middle of the Cold Aisle, Hot Aisle and Room	$C_{rap} = 4.6$ Symmetry

to simulate a medium-density rack with 9.8 kW/Rack and 9.8 °C delta T. This value is used based on the study by Fulpagare et al. [48] which used 20 servers operating at 490 W load each (maximum server load with the only CPU); (2) Hot aisle containment is deployed to isolate the hot aisle, but 5 cm under-rack gap and 3 cm in-rack leakage path are created to simulate the possible by-pass or recirculation air. The leakage path is critical to be included in the simulation as it bridges the pressure environment between the cold and hot aisle. [12,49,50]; (3) The room is set at 3 m high which is the same height as the FWC's specification [51], and a 0.5 m high return air plenum is added to create the return path to FWC unit. The FWC system is 2.4 m × 7.2 m × 3 m (Width x Length x Height); (4) the inlet and outlet pressures are set to 70 Pa and −60 Pa static, respectively. These values are taken from the pressure relationship in a data center hall reported by Joshi et al. [22]. The pressure drop from the return air plenum is modeled as the porous jump surface with a pressure drop coefficient estimated from 16 racks' design flowrate (29, 279 CFM), 90 Pa pressure drop [22] and return air plenum cross-section area 2.4 m² using a 2nd order polynomial in equation (1), resulting in

$$C_{rap} = \frac{90^2 \cdot 2}{\left(\frac{29279}{2118.88 \cdot 2.4} \right)^2} = 4.6.$$

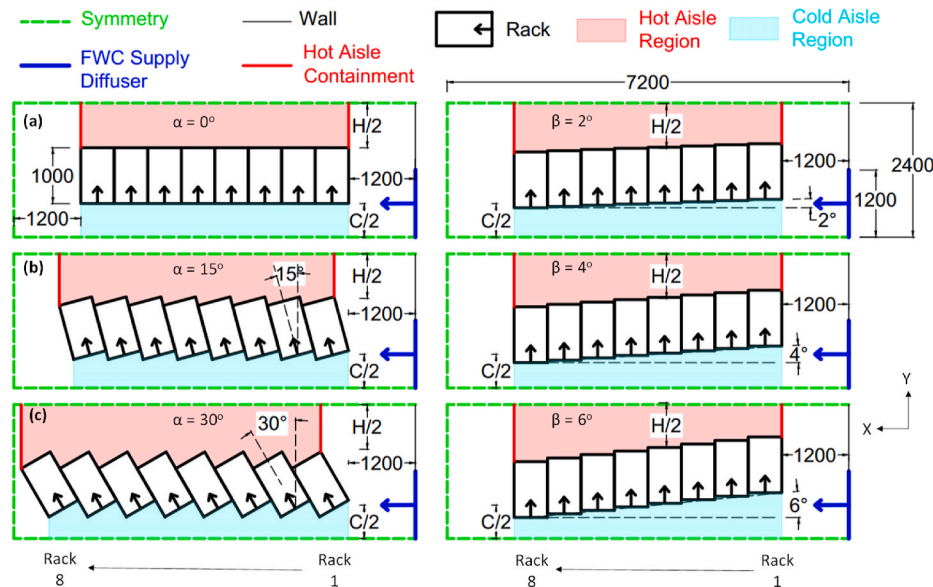


Fig. 5. Rack-Tilted (α) and Row-Tilted (β) Configurations (a) $\alpha = 0^\circ$; (b) $\alpha = 15^\circ$; (c) $\alpha = 30^\circ$; and (d) $\beta = 2^\circ$; (e) $\beta = 4^\circ$; (f) $\beta = 6^\circ$. The diagram shows half of a cold aisle and a hot aisle.

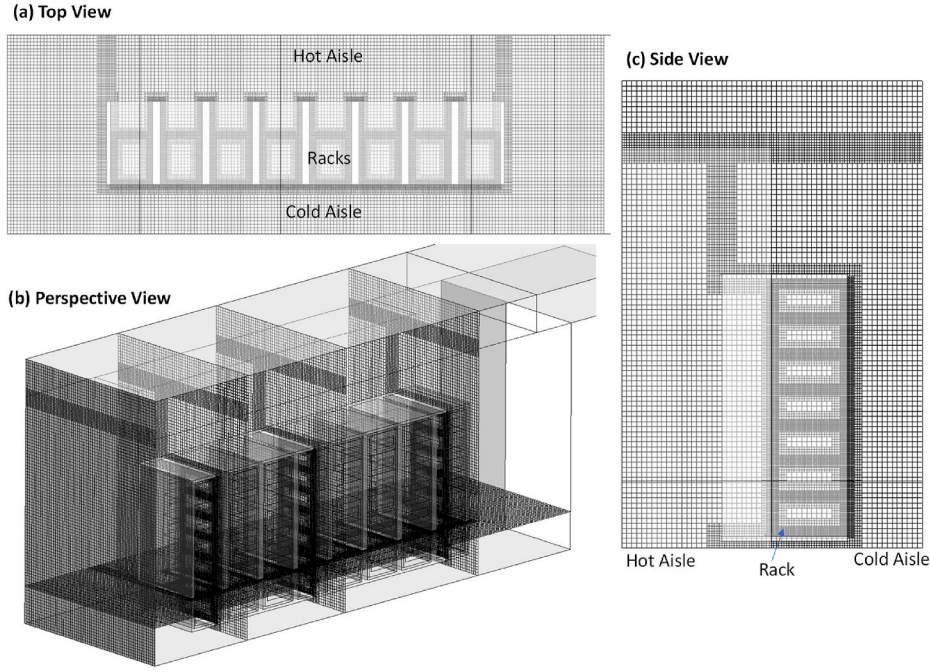


Fig. 6. FWC meshing (a) Top View; (b) Perspective View; (c) Side View.

Fig. 4 shows the geometry of the FWC numerical model with half a domain. It is worth highlighting that there are many possible FWC layouts, such as the one commonly seen in the cryptocurrency mining center supplying the air directly to each server cabinet [52]. The FWC in this study is a more compact version providing the air from the sidewall, having one FWC unit to handle the heat load of two rows of racks. A similar layout can be found in Ref. [42]. The full domain consists of two rows, and each row has eight racks. The simulation model treats the middle of the cold aisle, hot aisle and center of the room as symmetry to reduce the mesh size. The length of the room was set at 7.2 m. The FWC supply diffuser was placed 1.2 m away from the rack-row's leading edge (rack 1). According to the ASHRAE Thermal Guideline [53], the benchmark configuration was designed with eight tile distances from the cold to cold aisle. This translates to 4.8 m wide room consisting of 1.2 m cold aisle + 2 m rack + 1.6 m hot aisle. Table 3 summarizes the FWC boundary conditions. The model explores four spatial dimensions, Cold Aisle Width (C), Hot Aisle Width (H), Rack Titled Angle (α) and Row-Titled Angle (β). Six models were constructed as shown in Fig. 5.

The rack-tilted configuration rotates the rack with respect to the rack inlet center axis where the arrow labeled in the rack starts (Fig. 5). The rest of the racks was replicated in the same manner and placed horizontally. The row-tilt configuration offsets the rack with a certain distance one by one so that the entire rack row can have a tilted angle with respect to the horizontal line. Simple geometry calculations were performed to shift the tilted rack up or down to ensure all configurations have the same cold aisle area, as highlighted in Fig. 5.

2.3. Numerical method

The simulations were performed using Ansys Fluent 19.0. Incompressible 3D Reynolds-Averaged Navier-Stokes equations were solved using the finite volume method. Realizable k- ϵ turbulence model was selected referring with the earlier data center related simulations [29, 35, 54]. It is worth mentioning that the selection of the turbulence model could affect the accuracy of the simulation. Since the airflow in the DC is complicated and there is no universal turbulence model that can suit all flow scenarios. There are studies [14, 55] investigating the influence of turbulence models. Assuming the steady-state airflow, Newtonian fluid with constant density, viscosity and thermal conductivity at 300 K air

temperature, the continuity, momentum and energy equations are:

$$\frac{\partial u_i}{\partial x_i} = 0 \quad (2)$$

$$\frac{\partial}{\partial x_j} (u_i u_j) = -\frac{1}{\rho} \frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\nu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \overline{u_i' u_j'} \right] + \beta (T - T_0) g_i + S_i \quad (3)$$

$$\frac{\partial}{\partial x_j} (T u_j) = \frac{\partial}{\partial x_j} \left[\frac{\nu}{Pr} \frac{\partial T}{\partial x_j} - \overline{T' u_j'} \right] \quad (4)$$

where the turbulent stress and turbulent heat flux are

$$-\overline{u_i' u_j'} = \nu_t \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{2}{3} \left(k + \nu_t \frac{\partial u_k}{\partial x_k} \right) \delta_{ij} \quad (5)$$

$$-\overline{T' u_j'} = \frac{\nu_t}{Pr_t} \frac{\partial T}{\partial x_j} \quad (6)$$

The kinematic turbulent viscosity is calculated as

$$-\nu_t = C_\mu k^2 / \epsilon \quad (7)$$

where k and ϵ are the turbulence kinetic energy and turbulence kinetic energy dissipation rate, respectively. They are calculated through the transport equation:

$$\frac{\partial}{\partial x_i} (k u_i) = \frac{\partial}{\partial x_i} \left[\left(\nu + \frac{\nu_t}{\sigma_k} \right) \frac{\partial k}{\partial x_i} \right] + \frac{G_k}{\rho} + \frac{G_b}{\rho} - \epsilon \quad (8)$$

$$\frac{\partial}{\partial x_j} (\epsilon u_j) = \frac{\partial}{\partial x_j} \left[\left(\nu + \frac{\nu_t}{\sigma_\epsilon} \right) \frac{\partial \epsilon}{\partial x_j} \right] + C_{1\epsilon} S_\epsilon - C_{2\epsilon} \frac{\epsilon^2}{k + \sqrt{\nu \epsilon}} + C_{1\epsilon} \frac{\epsilon}{\rho k} C_{3\epsilon} G_b \quad (9)$$

where G_k is the generation of turbulence kinetic energy due to the mean velocity gradients. G_b is the generation of turbulence kinetic energy due to buoyancy. Except for the coefficient C_1 and C_μ are calculated from another two equations, the rest coefficients are constant, $C_{1\epsilon} = 1.44$, $C_{2\epsilon} = 1.9$, $\sigma_k = 1.0$, $\sigma_\epsilon = 1.2$.

$$C_1 = \max \left[0.43, \frac{\eta}{\eta + 5} \right], \eta = S \frac{k}{\epsilon} \quad (10)$$

Table 4

CFD Mesh Independent Study at benchmark configuration ($\alpha = 0^\circ$) at $C = 1.2$ m, $H = 1.6$ m

Global Size (cm)	Elements	Max Rack Inlet Temperature ($^\circ\text{C}$)	Max Rack Outlet Temperature ($^\circ\text{C}$)	Total Rack Flowrate (CFM)	Total Supply Flowrate (CFM)
3	8,384,253	27.5	35.3	28750	37601
4	4,334,085	27.3	35.1	28744	37608
5	2,683,575	28.9	36.1	28755	37615
6	1,706,873	28.1	36.4	28770	37587
7	1,237,913	30.7	37.6	28761	37582

Table 5

Comparison of the OHAS, FWC and UFAD.

	OHAS	FWC	UFAD
Domain Pressure Difference (Pa)	130	130	130
Supply Air flowrate (CFM)	29580	37608	26037
CRAC Fan Power	Medium	Low	High
Flow Maldistribution ($\Phi_{\max}$)	11%	14%	4%
Max Normalized Server Temperature ($T_{\text{ser,max}}^*$)	1.50	1.30	1.53
Installation	Relatively easier than UFAD as both OHAS and FWC have similar components such as the return air plenum. The only difference is the selection of the CRAC unit.	Complicated, the raised floor plenum involves many extra works	
Expected thermal inertia during cooling failure	Medium, the overhead plenum supplying the cold air could contribute the cooling a moderate duration of time.	Low, little solid structure exists in the supply channel.	High, the water pipe, pedestal and other solid elements in the raised floor plenum could support a longer operation.

$$C_\mu = \frac{1}{A_0 + A_s \frac{kU^*}{\varepsilon}} \quad (11)$$

where. $U^* \equiv \sqrt{S_{ij}S_{ij}}$, $A_0 = 4.04$, $A_s = \sqrt{6}\cos\varphi$, $\varphi =$

$$\frac{1}{3}\cos^{-1}\left(\sqrt{6}\frac{S_{ij}S_{jk}S_{ki}}{\sqrt{S_{ij}S_{ij}}}\right), S_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)$$

Pressure-based segregated algorithm with SIMPLE pressure-velocity coupling was chosen. The under-relaxation factor for pressure, body-

force, momentum, energy and turbulence were set to 0.3, 0.1, 0.2, 1.0 and 0.8 respectively. Second-order upwind spatial discretization was set for momentum, energy and turbulence. The standard wall function was used to approximate the velocity and shear stress in the log-law region. The Boussinesq approximation was used to capture the buoyancy effect of the airflow. The convergence criteria were set to 10^{-4} for the residuals of continuity, velocity and turbulent quantities, and 10^{-8} for the energy. The air properties were extracted from 300 K air temperature ($\rho_a = 1.16$ kg/m³, $\kappa_a = 0.0261$ W/mK, $\mu_a = 1.84 \times 10^{-5}$ kg/ms, $c_{p,a} = 1005$ J/kgK) with the solid materials in the rack to be aluminum.

The mesh was generated by Ansys Icepak 19.0 using the Hex-Dominant meshing method, which was economical and accurate for the room-scale simulations [29,56]. The local refinement uses the multi-level approach, which divides the globe mesh into multiple smaller size depending on the user input. The load bank, perforated tile, pressure inlet, and outlet had two refinement levels, while the rest of the objects with one level, as shown in Fig. 6. The mesh independent study explores five possible mesh sizes for the FWC study, as shown in Table 4. Eventually, the global mesh size 4 cm is chosen for further analysis (see Table 5).

2.4. Data reduction

In order to effectively evaluate the FWC performance, the normalized server temperature, T_{ser}^* is extracted from the previous study. It is a consolidated temperature weighted from the key server components from the reference 1U server [44]. Fig. 7 shows the normalized server temperature relationship to the normalized flowrate (V^*) and normalized rack inlet air temperature ($T_{a,\text{in}}^*$). V^* is calculated from simulated LB flowrate divide by its free delivery (FD) flowrate ($V_{\text{LB},0}$) as shown in equation (12). The free delivery LB flowrate at the high IT load is 261 CFM. $T_{a,\text{in}}^*$ is calculated from the simulated LB inlet air temperature $T_{a,\text{in}}$ divide the supply air temperature $T_{a,s}$ (25 $^\circ\text{C}$ for all cases) shown in equation (12). After obtaining V^* and $T_{a,\text{in}}^*$, T_{ser}^* is calculated using formula as shown in Fig. 7. This will bridge the relationship between the LB (a heater box) with a real server so that the results obtained from the LB performance could reflect the real server condition.

$$T_{a,\text{in}}^* = \frac{T_{a,\text{in}}}{T_{a,s}}$$

$$V^* = \frac{V_{\text{LB}}}{V_{\text{LB},0}} \quad (12)$$

$$\Phi = \left| \frac{V_{\text{LB},i} - \sum_0^N V_{\text{LB},i}/N}{\sum_0^N V_{\text{LB},i}/N} \right| \quad (13)$$

The maldistribution factor (Φ) is calculated using the individual LB flowrate divide the averaged LB flowrate in the simulation, as shown in equation (13). A similar parameter was used by studies [57,58] to

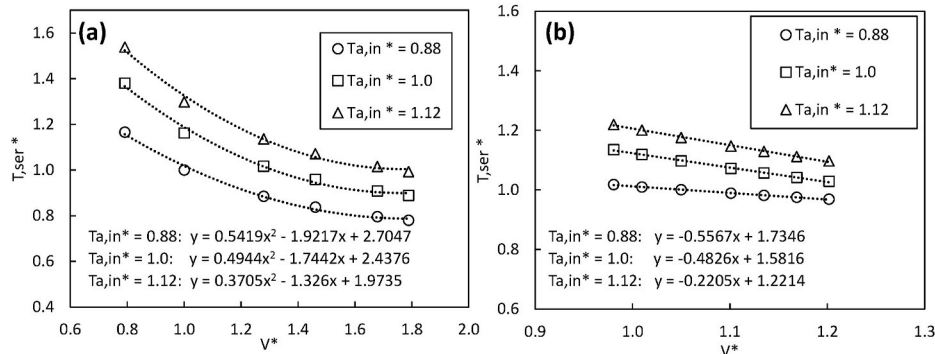


Fig. 7. Normalized Server Temperature Correlation (a) Low IT load, (b) High IT load.

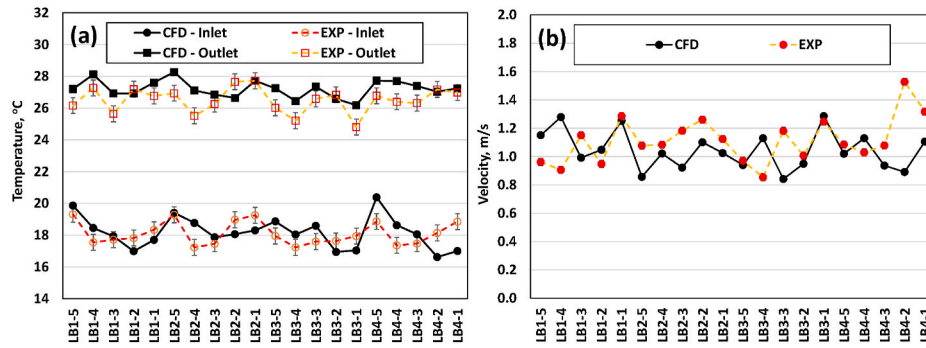


Fig. 8. Comparison of field measurement and CFD result (a) rack inlet and outlet temperature; (b) rack inlet velocity.

quantify the flow uniformity.

The total rack flowrate is defined as the summation of the LB air flowrates as shown in equation (14).

$$V_{rack} = \sum_{i=0}^N V_{LB,i} \quad (14)$$

The total supply flowrate is the mass-weighted volumetric flowrate in CFM (foot^3/min) under the fixed pressure difference boundary condition.

3. Results and discussions

3.1. Validation of the CFD model

Fig. 8 shows the comparison between the modeling and measurement results at each sensor point. The average rack inlet air temperature, $\bar{T}_{a,in}$ is 18.2 °C and 18.1 °C, the average rack out air temperature $\bar{T}_{a,out}$ is 27.2 °C and 26.5 °C in CFD and measurement, respectively. The LB temperature differences from both CFD and measurement are slightly lower than 9.8 °C since they are not mass-weighted averaged. Instead, they are calculated from the sensor points mounted at the rack inlet/outlet. The average rack inlet velocity $\bar{u}_a$ is 1.1 m/s and 1.0 m/s in CFD and measurement, respectively. It can be found that reasonable agreement has been achieved for most points, although some points deviate. It is also worth highlighting that the validation experiment compares a point air velocity obtained from the anemometer and CFD. The deviation may be attributed to (1) the rack model's simplification in CFD ignores the rack front mesh panel making the rack inlet air profile different; (2) simplification made to the perforated tile; (3) uncertain outflow direction from the domain. Overall, despite the uncertainties encountered during the field measurement, most of them are beyond the control because it is a live server room. The validation is considered valid as the model could accurately reflect the rack temperature difference and the velocity magnitude. Therefore, this model is translated and modified for the FWC study.

3.2. Comparison of the fan wall cooling (FWC), underfloor air delivery (UFAD) and overhead air supply (OHAS)

First, the FWC was compared with two commonly adopted air supply methods (i.e., the Underfloor Air Delivery and Overhead Air Supply). To have a fair comparison, all the boundary conditions in the UFAD and OHAS are identical with the FWC, except UFAD has a BF model to simulate tile air jet and OHAS has ceiling supply grills. The domain inlet (supply diffuser) and outlet have the same area. The UFAD supplies the cold air through the raised floor plenum and perforated tile, while the FWC supplies the cold air through side walls. The OHAS supplies the cold air through the ceiling plenum with grills opening (8 grills were modeled, each is 570 mm diameter). This size is twice bigger than the one in the study [41] due to a larger rack flowrate in the present study.

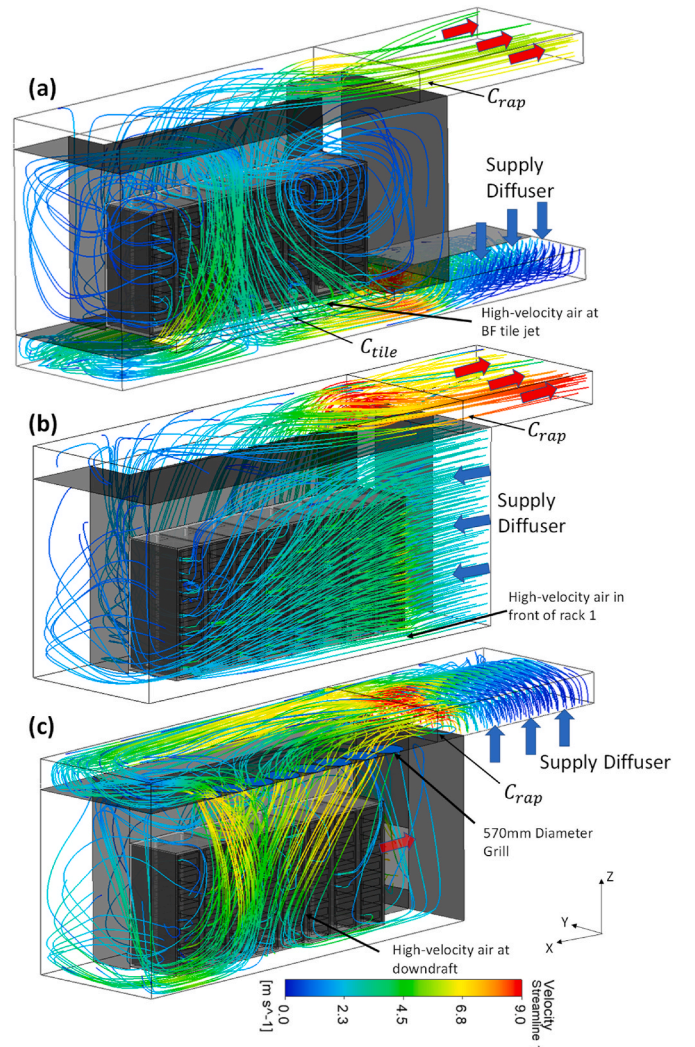


Fig. 9. Streamline of (a) UFAD; (b) FWC; (c) OHAS. All configurations are applied with hot aisle containment.

Fig. 9 depicts the streamline of all three configurations. Two noticeable recirculation regions were found in the front of rack 1, 2 and rack 7, 8 in UFAD and OHAS due to a bio-directional air jet from the perforated tile and ceiling grills. Similar patterns were observed in the previous studies [59,60]. The high-velocity air jet from the perforated tile in UFAD and downwards draft from ceiling grills in OHAS create the low static pressure zones that induce recirculation from the hot aisle. Nemati et al. [61] conducted flow visualization and provided concrete

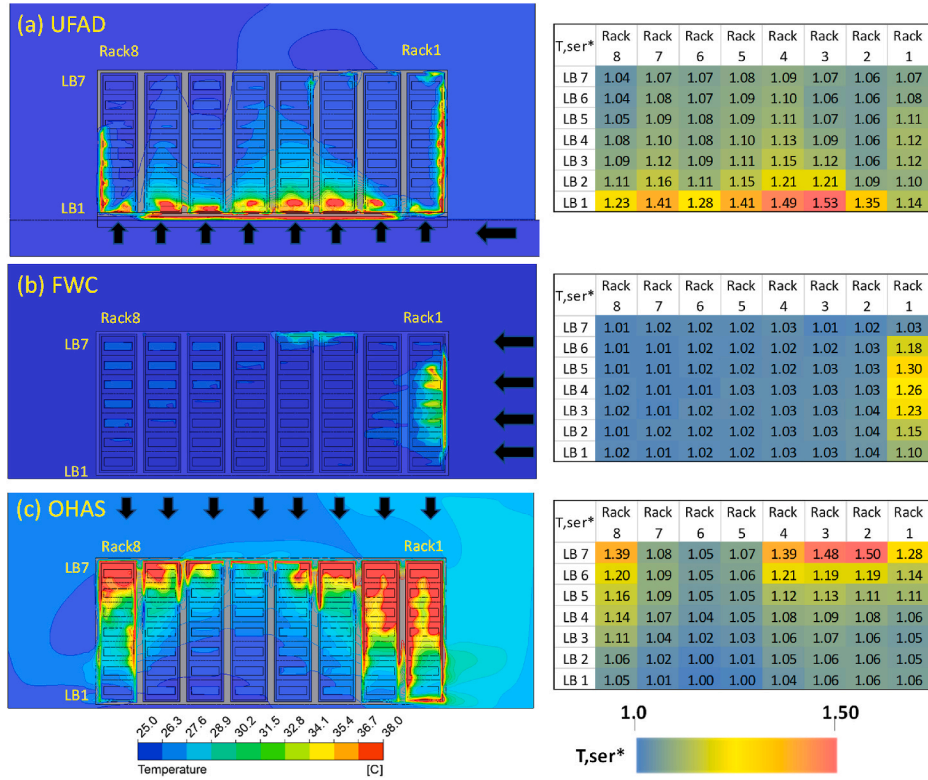


Fig. 10. Temperature Contour (Front View) in the cold aisle (0.1 m in front of racks), and the normalized server temperature of each Load Bank (LB) in the racks (a) UFAD; (b) FWC; (c) OHAS. All configurations are applied with hot aisle containment, the hot air recirculation occur through leakage path (under-rack, in-rack gap).

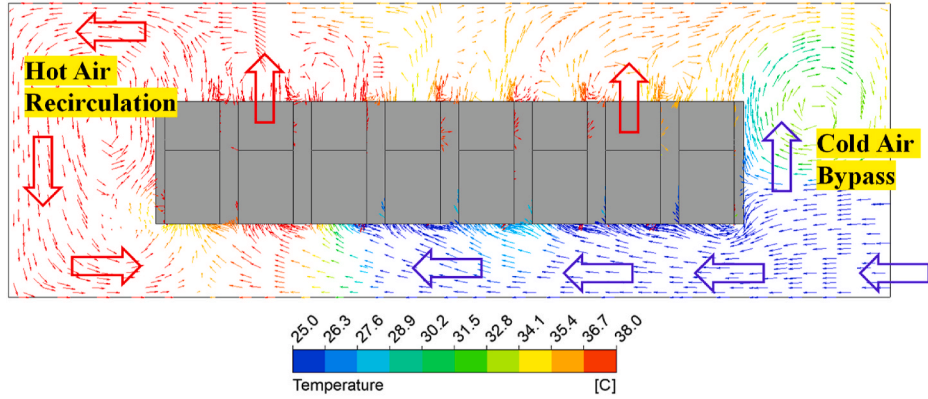


Fig. 11. Vector plot of open aisle FWC.

evidence of tile air-jet induced flow.

Fig. 10 shows the normalized server temperature distribution and temperature contour in front of the rack for each layout. The normalized server temperature considered the changes of the rack inlet air temperature as well as the rack air flowrate, as shown in Fig. 7. It can be found that the LB 1 located at the bottom of the racks in UFAD suffers a significant amount of recirculation air, elevating the maximum normalized server temperature ($T_{ser,max}^*$) to 1.53, 15% higher than FWC that has $T_{ser,max}^* = 1.30$. Similarly in OHAS, the $T_{ser,max}^*$ reaches 1.50 due to substantial recirculation induced by high-velocity air jet.

Under the same pressure difference boundary condition (dp_{room}), the FWC supplies 37608 CFM flowrate for 16 racks, while UFAD only supplies 26037 CFM. It is about 44.6% improvement due to the elimination of the perforated tile and raised floor plenum (C_{tile}). In OHAS, the ceiling grills were modeled as free-opening with no pressure drop coefficient assigned. Although they are big (twice as big as the [41]), it

creates a strong downwards impinging jet, the total supply flowrate in OHAS is 29580 CFM, lower than FWC but slightly higher than the UFAD. Therefore, the result of this section implies that the FWC has a better thermal performance than UFAD and OHAS.

It is worth noting that the performance of the FWC in open aisle configuration (i.e. no containments) is worse than the UFAD and OHAS due to excessive cold air by-pass and hot air recirculation, as shown in Fig. 11. In the next section, the influence of cold and hot aisle width in FWC with hot aisle containment will be analyzed.

3.3. Influence of cold aisle size (C)

FWC supplies cold air from the sidewall, making Rack 1 the most critical spot. Fig. 12 (a) shows the maldistribution factor at $\alpha = 0^\circ$ when hot aisle width (H) is fixed at 1.6 m while the cold aisle width (C) gradually increases from 0.6 m to 1.8 m with 0.3 m interval. It is

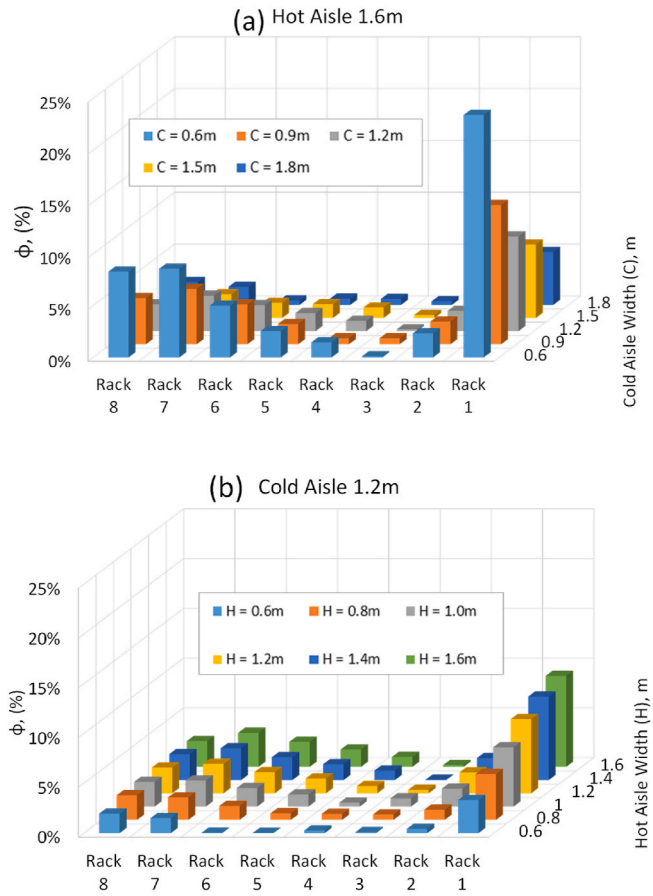


Fig. 12. Maldistribution Factor at $\alpha = 0^\circ$, (a) $H = 1.6$ m; (b) $C = 1.2$ m

noticeable that Rack 1 (leading edge) suffers the severest maldistribution among all racks due to the low static pressure zone created by the high-velocity air in front of Rack 1. This is because of the sudden shrinkage of the flow path at the cold aisle entrance. Increasing the cold aisle width will assist in relieving such maldistribution. The results in Fig. 12 (a) show that the maldistribution factor (ϕ) is gradually reduced from 23.4% at $C = 0.6$ m to 5% at $C = 1.8$ m.

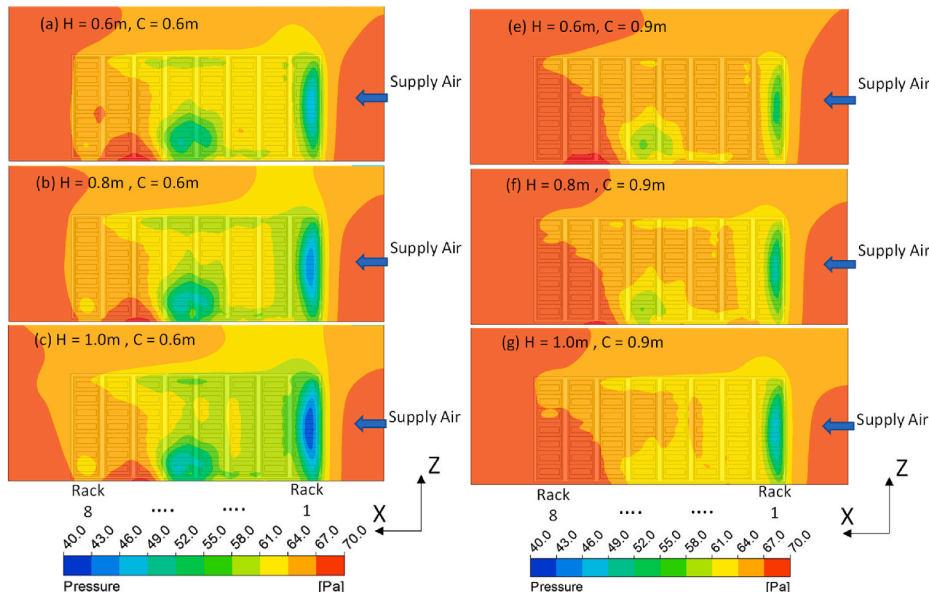


Fig. 13. Pressure Contour (Front View) at base case ($\alpha = 0^\circ$) in the cold aisle (0.1 m in front of racks).

Fig. 13 shows the pressure contour of each FWC at different cold aisle widths. The low-pressure zone in $C = 0.9$ m case is significantly smaller than $C = 0.6$ m. This results in much less recirculation air in $C = 0.9$ m, especially in front of Rack 1, as shown in Fig. 14. Table 6 shows that the average maldistribution factor ($\bar{\phi}$) gradually reduces from $\bar{\phi} = 6.5\%$ at $C = 0.6$ m to $\bar{\phi} = 1.5\%$ at $C = 1.8$ m. This suggests that enlarging the cold aisle width helps to mitigate the flow maldistribution and reduce the recirculation.

3.4. Influence of hot aisle size (H)

Fig. 12(b) depicts the maldistribution factor at $\alpha = 0^\circ$ when cold aisle width (C) is fixed at 1.2 m while the hot aisle width (H) gradually increases from 0.6 m to 1.6 m with 0.2 m intervals. It is interesting to find that the maldistribution gets severer with the enlargement of the hot aisle. The maldistribution factor increases from 3.3% at $H = 0.6$ m to 9.1% at $H = 1.6$ m. The pressure contour in Fig. 14 of $H = 0.6$ m, $H = 0.8$ m and $H = 1.0$ m confirms the observation. To explain the cause, it is worth examining a closure look at the flowrate distribution among the racks. Table 6 shows individual rack flowrate at different cold and hot aisle widths. The rack air flowrate is more evenly distributed at $H = 0.6$ m with the average maldistribution factor $\bar{\phi} = 1.0\%$, while it is most unevenly distributed at $H = 1.6$ m with the $\bar{\phi} = 2.8\%$. This suggests that a small hot aisle can help balance the pressure in the cold aisle by recirculating an additional volume of air back to the cold aisle through leakage gap to mitigate maldistribution and increases the cold aisle temperature. The leakage paths (under-rack and in-rack gaps) bridge the cold aisle and hot aisle, allowing such recirculation to occur. Comparatively, less recirculation appears in front of Rack 3 to 5 and more recirculation in front of Rack 1 with the increase of the hot aisle width (from the temperature contour in Fig. 14 of $H = 0.6$ m, $H = 0.8$ m and $H = 1.0$ m). This section's analysis implies that enlarging the hot aisle width to mitigate the flow maldistribution is not an ideal solution and needs further analysis for the FCW design.

3.5. Optimization of fan wall cooling (FWC)

As shown in Fig. 5, another five possible rack configurations ($\alpha = 15^\circ$; $\alpha = 30^\circ$; $\beta = 2^\circ$; $\beta = 4^\circ$; $\beta = 6^\circ$) are evaluated to mitigate the flow maldistribution and hot air recirculation in standard FWC design. Fig. 15 compares the maximum maldistribution factor (ϕ_{max}) among all racks in

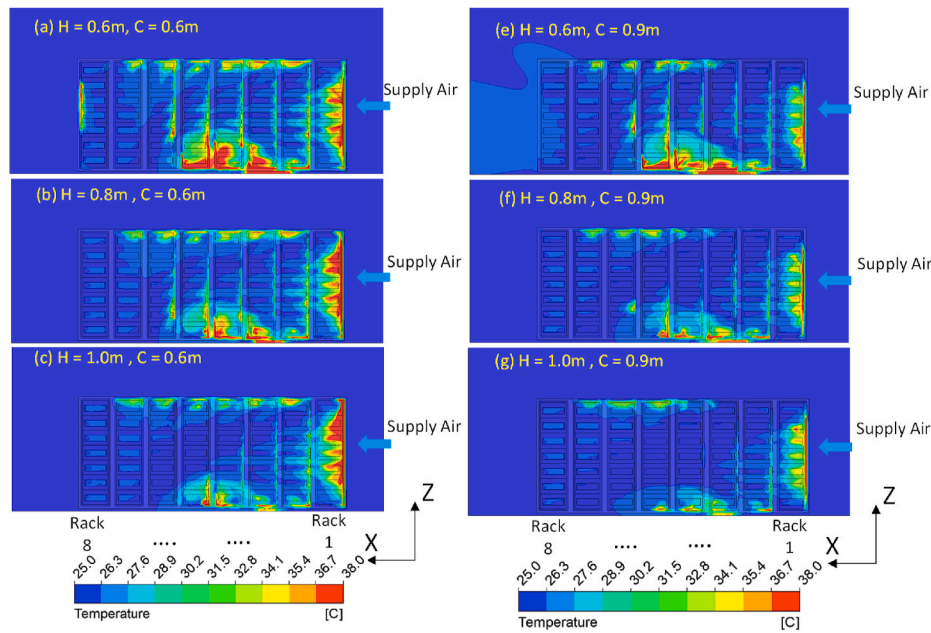


Fig. 14. Temperature Contour (Front View) at $\alpha = 0^\circ$ in the cold aisle (0.1 m in front of racks).

Table 6

Rack air flowrate at various Cold Aisle Width at fixed $H = 1.6$ m, and different Hot Aisle Width at fixed $C = 1.2$ m

Rack Air Flowrate (CFM)	Rack 8	Rack 7	Rack 6	Rack 5	Rack 4	Rack 3	Rack 2	Rack 1	$\bar{\phi}$
$C = 0.6$ m ($H = 1.6$ m)	1875	1880	1818	1775	1757	1729	1691	1326	6.5%
$C = 0.9$ m ($H = 1.6$ m)	1854	1869	1843	1809	1785	1764	1735	1536	4.1%
$C = 1.2$ m ($H = 1.6$ m)	1844	1858	1842	1828	1815	1794	1762	1633	2.8%
$C = 1.5$ m ($H = 1.6$ m)	1856	1861	1846	1843	1837	1824	1794	1690	2.1%
$C = 1.8$ m ($H = 1.6$ m)	1884	1875	1851	1854	1854	1849	1827	1749	1.5%
$H = 0.6$ m ($C = 1.2$ m)	1798	1790	1763	1763	1768	1766	1756	1705	1.0%
$H = 0.8$ m ($C = 1.2$ m)	1808	1804	1789	1776	1753	1754	1746	1683	1.7%
$H = 1.0$ m ($C = 1.2$ m)	1813	1815	1802	1790	1776	1754	1737	1663	2.2%
$H = 1.2$ m ($C = 1.2$ m)	1821	1828	1813	1802	1789	1769	1738	1643	2.5%
$H = 1.4$ m ($C = 1.2$ m)	1833	1844	1828	1815	1804	1786	1747	1637	2.7%
$H = 1.6$ m ($C = 1.2$ m)	1844	1858	1842	1828	1815	1794	1762	1633	2.8%

each configuration at the same cold aisle width ($C = 1.2$ m) while the cold aisle changes from $H = 0.6$ m– 1.6 m. Fixing the cold aisle width for optimization is due to the difficulty of compressing the cold aisle since it needs sufficient space (1.2 m wide) for regular maintenance.

It can be seen that the maldistribution is the severest in the benchmark configuration ($\alpha = 0^\circ$) with $\phi_{\max} = 9.1\%$ in $H = 1.6$ m. The trends show that the maldistribution is gradually softened in the new configurations when the rack or rack-row is tilted. The $\phi_{\max}$ is reduced to 4.0% in $\alpha = 30^\circ$ configuration at same hot aisle width ($H = 1.6$ m). Such improvements have been observed because (1) the rack-tilted configurations ($\alpha = 15^\circ$; $\alpha = 30^\circ$) have a better alignment with the approaching supply air in the cold aisle as shown in the vector plot in Fig. 17. (2) the row-tilted configurations ($\beta = 2^\circ$; $\beta = 4^\circ$; $\beta = 6^\circ$) have an enlarged cold aisle entrance, reducing the supply air velocity in front of rack 1.

Fig. 15 (b) shows that the maximum rack inlet air temperature ($T_{a,in,max}$) increases in all configurations when the hot aisle is shrinkage from $H = 1.6$ m to $H = 0.6$ m. It can be found that the $T_{a,in,max}$ reduces when switching from the benchmark to rack tilted $\alpha = 15^\circ$ configuration. Since the supply air temperature from FWC is fixed at 25°C , the higher the $T_{a,in,max}$ means more recirculation has occurred. Except for the new configuration $\alpha = 15^\circ$ has a smaller $T_{a,in,max}$, the rest has a higher value in general, especially for row-tilted configurations ($\beta = 2^\circ$; $\beta = 4^\circ$; $\beta = 6^\circ$). This is because the enlarged cold aisle entrance at Rack 1 in row-tilted configurations reduces hot aisle width at the rear of Rack 1, inducing additional recirculation air.

Fig. 15 (c) and (d) show the total rack air flowrate and FWC supply air flowrate from $H = 0.6$ m to $H = 1.4$ m in new and benchmark configurations. Since the rack was assigned with an active fan curve, it responds to preferable or adverse pressure difference. For instance, when no external pressure difference exists between the front and rear of the rack, it is known as the Free Delivery (FD), which is labeled as a dotted line in Fig. 15 (c). When a preferable pressure difference (i.e., provision of a cold aisle containment has a higher pressure at rack's intake), the rack flowrate will increase. Vice versa, the hot aisle containment adopted in this study creates an adverse pressure difference, reducing the rack air flowrate, as shown in Fig. 15 (c). This explains why all racks have a flowrate lower than its FD condition. It can be seen that the rack flowrate increases 800 CFM when switches from the benchmark ($\alpha = 0^\circ$) to rack-tilted ($\alpha = 15^\circ$) configuration. And all other new configurations have a higher rack flowrate than the benchmark ($\alpha = 0^\circ$). As shown in vector plot Fig. 17 (b). The rack intake is perpendicular to the supply air in the benchmark ($\alpha = 0^\circ$), and the new configurations have a smaller angle with supply air after tilting. Such adjustments allow more air to flow towards the rack and increases rack air flowrate. However, the formation of the vortices in front of the rack can reduce the rack air flowrate if the racks are tilted too much (i.e., $\alpha = 30^\circ$). Therefore, tilting the rack to 15° is the most appropriate in new FWC configurations.

In the end, the maximum normalized server temperature ($T_{ser,max}^*$) is shown in Fig. 16. The increased rack air flowrate and reduced recirculation in rack-tilted $\alpha = 15^\circ$ configuration enhance the server level

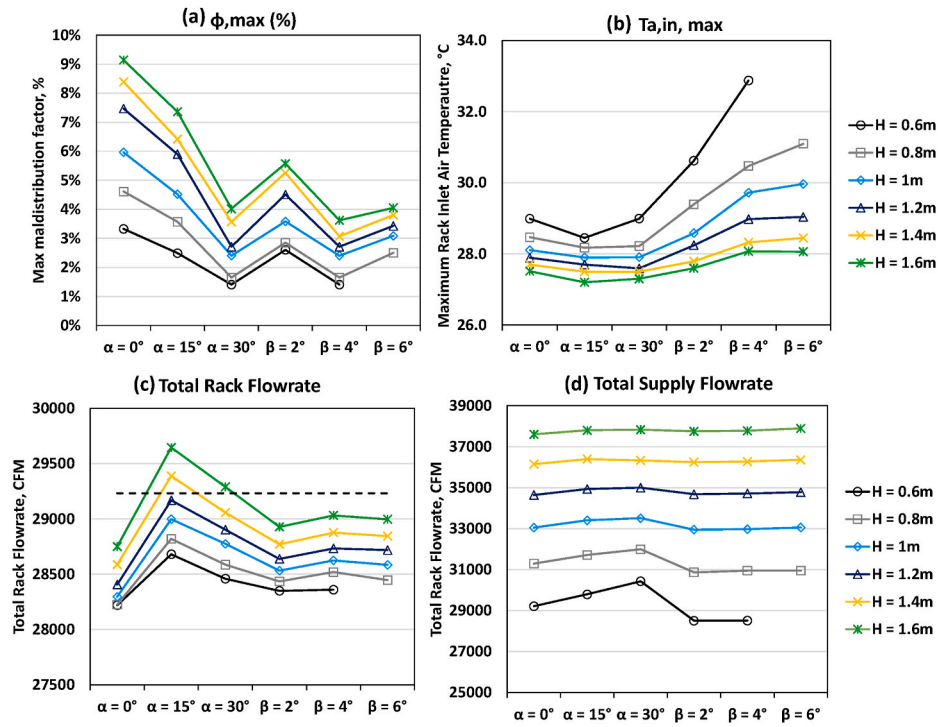


Fig. 15. At cold aisle width, $C = 1.2\text{ m}$ (a) Maximum maldistribution factor; (b) Maximum rack inlet air temperature; (c) Total Rack Flowrate; (d) Total Supply Flowrate.

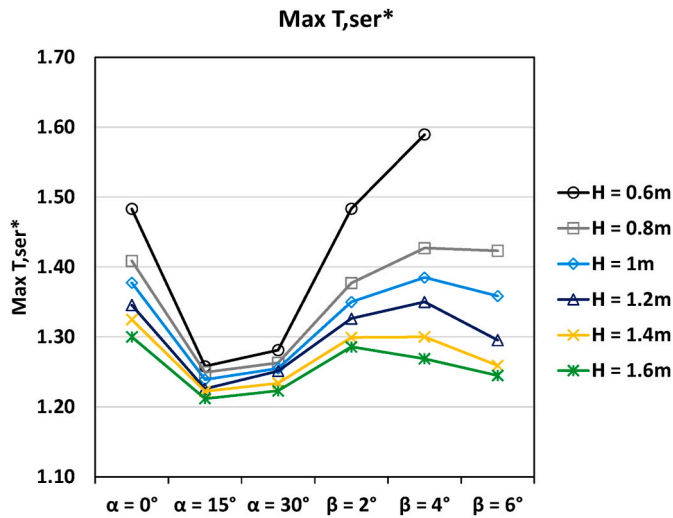


Fig. 16. Maximum Normalized Server Temperature at cold aisle width $C = 1.2\text{ m}$

thermal performance. The $T_{ser, \max}^*$ reduces from 1.30 in benchmark ($\alpha = 0^\circ$) to 1.21 in rack-tilted $\alpha = 15^\circ$ configuration. It also shows that adopting the row-tilted configurations ($\beta = 2^\circ$; $\beta = 4^\circ$; $\beta = 6^\circ$) could improve the server thermal performance as well if the hot aisle is wide enough (i.e., hot aisle width $H > 1.2\text{ m}$).

4. Conclusion

We presented six Fan Wall Cooling (FWC) configurations and studied the airflow and thermal performance for various conditions. Although the exhaust air profile of the load bank is more diffuse than the server, their responses to the changes of the environment (such as the back-pressure in hot aisle containment) are similar. Therefore, using server

temperature correlation could effectively bridge the load bank performance with an actual server for evaluation. Through the validated CFD model, the advantage of the FWC over Underfloor air delivery (UFAD) and Overhead air supply (OHAS) have been highlighted. However, the FWC does have its drawback due to flow maldistribution at the cold aisle entrance. The proposed new configurations effectively eliminate this drawback by tilting the rack with an angle. The conclusions of the current work are summarized as below:

- Comparing the FWC with the UFAD and OHAS, the maximum normalized server temperature ($T_{ser, \max}^*$) is 0.19 and 0.63 lower than UFAD and OHAS, respectively. The elimination of the raised floor plenum and perforated tile increases the supply air flowrate by 44.6% from 26037 CFM to 37608 CFM
- In fan wall cooling, the flow separation and the low static pressure zone at the rack-row's leading-edge creates hot spot by inducing excessive recirculation from the hot aisle. Maldistribution factor ($\phi_{\max}$) can reach up to 23.4% for the smallest cold aisle width. Enlarging either the cold or hot aisle width improves the airflow maldistribution and reduces the recirculation.
- The proposed rack-tilted and row-tilted FWC configurations could effectively mitigate the flow maldistribution and improve the normalized server temperature T_{ser}^* . Maldistribution factor ($\phi_{\max}$) is reduced from 7% to 23.4% for the benchmark configuration to 1.5%–5.1% for $\alpha = 30^\circ$. The optimum results show that the rack air flowrate increases 2.5% (800 CFM), and normalized maximum server temperature ($T_{ser, \max}^*$) reduces by 7% from 1.30 to 1.21 in the optimum FWC configuration ($\alpha = 15^\circ$).

CRedit authorship contribution statement

Xin Xiong: Conceptualization, Methodology, Software, Validation, Investigation, Writing - original draft, Writing - review & editing. **Yogesh Fulpagare:** Experimentation, Validation, Resources. **Poh Seng Lee:** Conceptualization, Supervision, Funding acquisition, Resources.

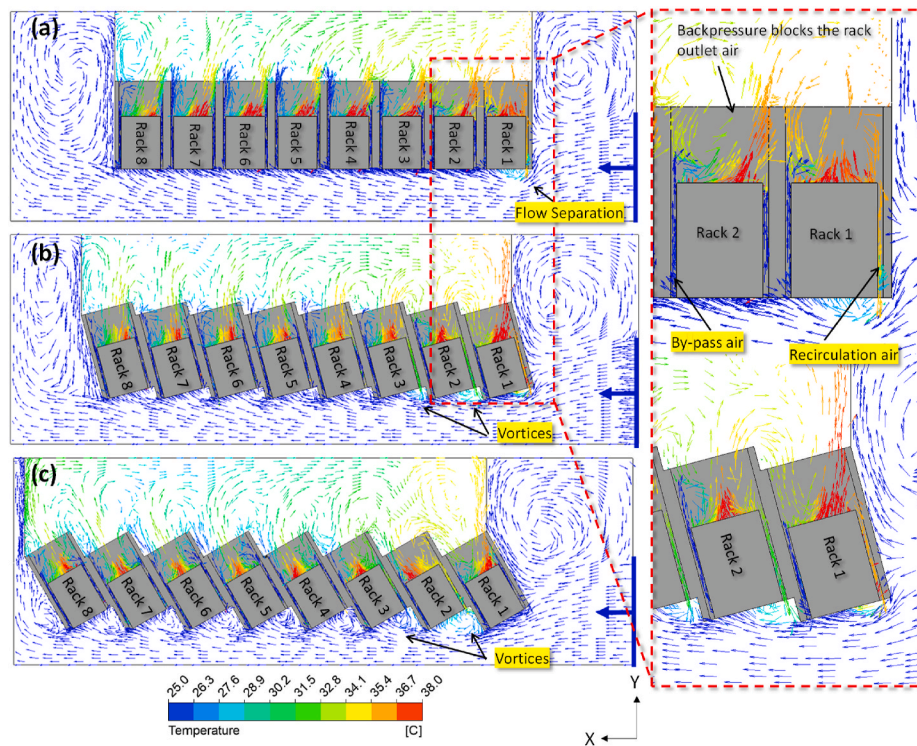


Fig. 17. Vector Plot (top view, colored with temperature, at x-y plane 1.5 m above the floor) at $C = 1.2$ m, $H = 1.6$ m, (a) $\alpha = 0^\circ$; (b) $\alpha = 15^\circ$; (c) $\alpha = 30^\circ$.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Any opinions, findings and conclusions or recommendations expressed in this material are those of the author(s) and do not reflect the views of National Research Foundation (NRF) Singapore and Infocomm Media Development Authority (IMDA) of Singapore.

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